Making Organizational Behavior Observable:
An Auditable Multimodal Infrastructure for Measuring Behavior from Audio and Video

Abstract

Organizational interactions are increasingly recorded, yet most recordings remain difficult to use as scientific data. Transcripts discard vocal delivery, conversational timing, visible behavior, and production context. Audio and video preserve this information, but researchers still need a way to organize it into measures linked to the original recording. We develop an AI-assisted measurement infrastructure that converts heterogeneous recordings into time-aligned, source-linked evidence and enables researchers to construct and inspect multimodal measures across settings. We define behavior as research-observable when it can be measured transparently and traced to a person and moment in the original recording. We demonstrate the infrastructure using audio from 286 earnings calls involving 12 executives and video from 30 public interviews and eight additional interviews with the same interviewer. The pilots show that apparently routine language in transcripts can accompany unusual vocal delivery or timing, and that words, voice, and visible behavior do not always follow the same pattern. AI helps extract and organize observations at scale; researchers decide what to measure, how to validate it, and what conclusions the evidence supports. By linking measures to the original recordings and documenting how they were produced and where evidence is uncertain or missing, the system helps researchers check, revise, and adapt measures across settings.

Organizational interactions are increasingly recorded, yet most recordings remain difficult to use as scientific data. Transcripts discard vocal delivery, conversational timing, visible behavior, and production context. Audio and video preserve this information but do not make it structured, comparable, or traceable. We develop an AI-assisted measurement architecture (intended for open-source release) that turns heterogeneous recordings into time-aligned, source-linked evidence and enables researchers to construct, inspect, revise, and apply multimodal measures across settings. Behavior becomes research-observable when researchers can locate it in the source, link it to a person and moment, reconstruct its measurement, and identify missing evidence. AI supports extraction and organization; researchers define constructs, validate measures, and draw inferences.

1. From recorded to research-observable behavior

Organizational and social scientists seek to explain how people communicate, influence one another, and coordinate action. These processes unfold across words, vocal delivery, timing, visible behavior, and responses from others (LeBaron et al., 2018; Lehmann-Willenbrock, 2025). Yet researchers often rely on transcripts, surveys, or downstream outcomes because extracting and aligning multimodal information from recordings remains costly. Even when recordings are analyzed, moment-to-moment interaction is often compressed into a few static variables. This makes theories of organizational behavior difficult to test across naturally occurring interactions at scale.

The problem takes three different forms. First, *channel loss* occurs when information from one or more channels is discarded. A transcript, for example, preserves words but not how they were spoken, when a response began, whether speakers overlapped, or how others visibly reacted. Second, *measurement friction* occurs when behavior remains audible or visible but is too costly to structure and compare across many recordings. Third, *source non-observability* occurs when the recording never captured the relevant behavior—for example, because a listener was off-camera or a face was obscured. Separating these three problems makes clear both what AI-assisted measurement can recover and what it cannot.

AI tools can help address the first two problems. Speech recognition, identifying who is speaking and when, and computer vision can recover information omitted from transcripts and make it easier to measure across recordings. They cannot recover behavior that a recording never captured, but can mark it as unavailable. Extracting a pause, pitch change, smile, or gaze shift also does not establish emotion, intent, or deception (Russell et al., 2003; Barrett et al., 2019). Researchers must still define constructs and validate how observations are combined and interpreted, while recognizing that model choices can affect the results (Harvey et al., 2026; Carlson & Burbano, 2026). Our aim is to extract observable information, identify what is missing, and leave theoretical interpretation to researchers.

2. What exists and what we add

Existing research provides the building blocks for this project. Organizational researchers have developed theories and coding methods for speech, facial movement, gaze, turn-taking, and interaction sequences, while automated tools can extract many of these features (Baltrušaitis et al., 2018; Liu et al., 2023). Multimodal corpora and studies of teams and earnings calls show the value of combining language, voice, visible behavior, and context (Reece et al., 2022; Lehmann-Willenbrock & Hung, 2024; Mayew & Venkatachalam, 2012; Rennekamp et al., 2022; Baik et al., 2025). What is still missing is a reusable way to turn outputs from different tools into measures that researchers can inspect, revise, validate, and test across settings.

Our contribution is a measurement architecture that turns observations from recordings into measures researchers can inspect and revise. It uses existing tools to identify speakers and conversational turns and extract language, timing, vocal delivery, camera cuts, participant visibility, and facial movement while speaking and listening. The architecture aligns these observations by person and time and links them to

the original recording, so researchers can check the evidence behind each measure. It has two linked layers (Figure 1).

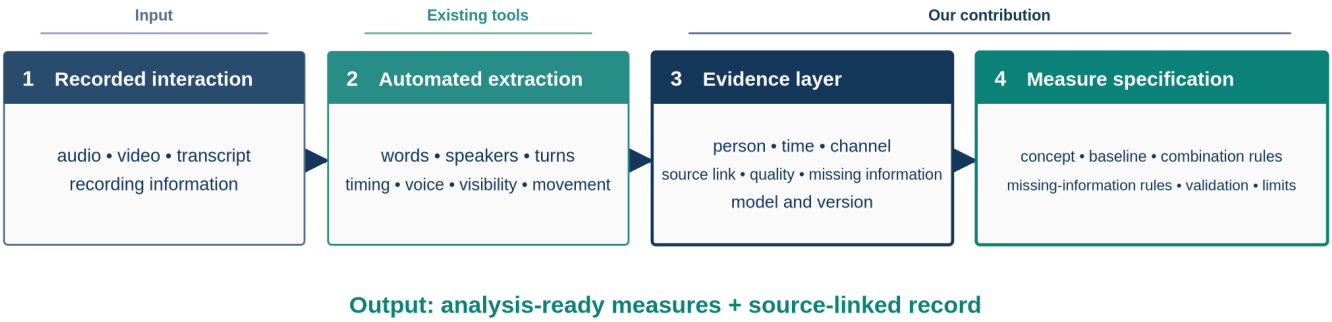


Figure 1. The measurement system. *Existing tools extract information from recordings. Our architecture aligns their outputs by person, time, and channel, links them to the source, and allows researchers to specify how observations form measures for analysis. Extraction, measurement, and interpretation remain separate.*

The first is an evidence layer that aligns observations across people, time, and channels. It records what was observed, where it came from, whether it was available, any quality concerns, and which model and version produced it. This prevents an off-camera participant from being coded as showing no facial movement or an uncertain speaker assignment from being treated as correct. Keeping the observations aligned also preserves interaction sequences rather than reducing behavior to isolated features (Knox & Lucas, 2021).

The second is a measure-specification layer that records how observations become variables. Researchers specify the construct, observations, comparison baseline, treatment of missing information, and validation plan. For example, a pause remains a measured duration and a departure from a stated baseline; it is not automatically labeled uncertainty. Researchers can inspect the source, revise rules or baselines, and compare alternative measures. A short narrated worked example follows one executive answer from source audio and transcript to component measurements, robustness checks, and an exported observation, illustrating how researchers can inspect the evidence underlying a measure.[1]

3. Prototype and initial evidence: two pilot applications

We tested the prototype in two settings that reveal different limitations of conventional data. The earnings-call pilot examines whether audio can recover departures that transcripts miss. The video-interview pilot examines how camera choices affect what behavior can be observed. These pilots demonstrate what the architecture can do; they do not establish that the observed departures measure any particular theoretical construct.

Earnings-call pilot. We collected 300 calls involving 12 executives across seven sectors. Of these, 286 calls met the analysis requirements, representing approximately 313 hours of audio and producing 3,273 executive answers during live Q&A. We established a separate baseline for each executive from their answers across calls and compared each answer with that executive's usual language, vocal delivery, and interaction timing.

We identified 177 answers in which the language was relatively typical for that executive but the vocal delivery or interaction timing was unusual (Figure 2). Specifically, these answers had a multimodal

[1] The project repository contains the prototype, derived data, figures, technical appendix, and researcher guide: https://github.com/heeyonkim/auditable-multimodal-measurement.

departure score of at least 2.0 and a transcript departure score of no more than 1.25. We treat them as cases for further investigation, not as evidence of emotion, intent, truthfulness, or executive quality. The finding did not depend on one type of audio information. After separately excluding timing and speaking rate, pitch, or vocal intensity, 172 of the 177 answers remained unusual in at least two of the three analyses. The measurement system retains the continuous scores so researchers can inspect the degree and source of each departure rather than treating a cutoff as a fact.

Our transcript comparison used answer length, filled pauses, discourse fillers, and adjacent repetitions; it did not capture semantic content or all information available from text. At a departure-score cutoff of 2.0, the audio/interaction measure flagged 325 answers and the transcript-feature measure flagged 321, with 66 flagged by both. Thus, 259 of the 325 answers (80%) flagged by the audio/interaction measure were not flagged by this transcript-feature baseline. This does not establish which measure is correct, but shows that the two feature sets identify substantially different answers for further inspection.

To explore whether these departures relate to organizational outcomes, we linked call-level scores to market reactions. Across 274 calls with sufficient eligible Q&A answers, greater median departure from an executive's usual vocal delivery and interaction timing was associated with slightly lower market-adjusted returns, after accounting for transcript-visible departure, firm, and year (partial Spearman correlation = −.10, p = .094). This exploratory analysis demonstrates how the measures can be used in a study; it does not establish predictive validity or a causal relationship.

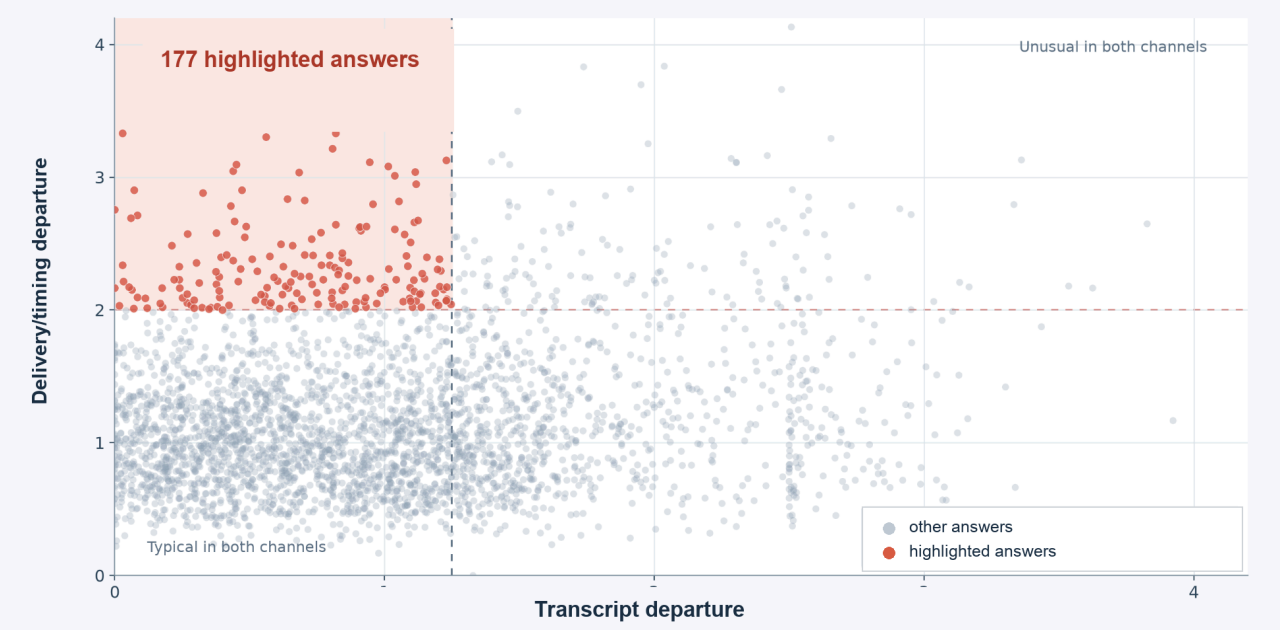


Figure 2. Audio surfaces departures that transcripts miss. *Each point represents an answer compared with that executive's own baseline. Highlighted answers were relatively typical in transcript language but unusual in vocal delivery or interaction timing.*

Video-interview pilot. Video preserves visible behavior that transcripts and audio omit, but it remains difficult to analyze systematically at scale. Researchers must align visible behavior with speech and interaction timing while accounting for how camera selection and editing affect what is observable (LeBaron et al., 2018; Lehmann-Willenbrock & Hung, 2024). We examined this problem using 30 Bloomberg interviews spanning 8.6 hours and several production formats. Across interviews, the median proportion of speaker transitions occurring within two seconds of a camera cut was 66%, compared with a median chance expectation of 17% based on randomly shifted transition times within the same videos. If editing is ignored, a researcher could mistake a production decision for a change in participant

behavior. The prototype automatically detects camera cuts and records participant visibility separately, treating periods when someone is not visible as unavailable rather than as an absence of facial movement.
In a separate sample of eight interviews from Bloomberg's Leaders with Lacqua, we compared interviewer Francine Lacqua's transcript features, turn-taking, vocal delivery, and visible speaking and listening behavior across encounters with different CEOs. Each encounter was compared with other eligible interviews. In the interview with CEO Federico Marchetti, her transcript features were close to her usual pattern, while turn-taking and vocal delivery differed (Figure 3).

A sensitivity analysis showed why the treatment of unavailable visual information matters. A simple approach that treated periods when Francine Lacqua was not measurably visible as showing no facial movement produced a different ranking from our production-aware approach, which treated those periods as unavailable and accounted for framing and camera cuts. Among the six interviews eligible under both specifications, the interview with CEO Pascal Soriot moved from fifth to first in measured distinctiveness when unavailable visual information, framing, and camera cuts were handled explicitly.

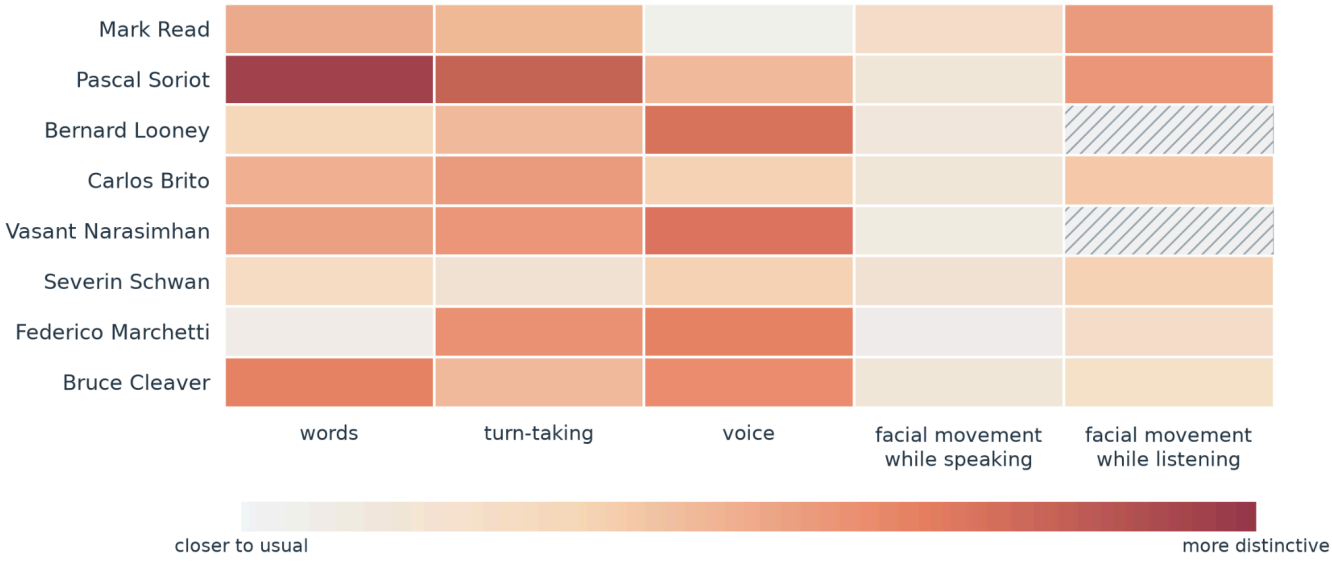


Figure 3. Behavioral patterns vary across channels. *Each row represents an interview and is labeled with the CEO interviewed; each column represents a channel. Color indicates how much the interviewer departed from her pattern across other eligible interviews: seven for words, turn-taking, voice, and facial movement while speaking, and five for facial movement while listening. Hatched cells indicate insufficient observable evidence to calculate a score.*

Together, the pilots demonstrate the feasibility of recovering information omitted from transcripts, comparing behavior within people, and distinguishing unavailable visual information from an absence of behavior.

4. Stage 2: From working prototype to reusable measurement system
Stage 2 will extend and validate the working prototype for use across a wider range of recordings and research settings. We will focus on four areas: technical accuracy, measure validation, researcher use, and empirical expansion.

Technical benchmarking. We will compare the system's identification of speakers, turns, timing, vocal features, camera cuts, visibility, and facial movement with human-reviewed examples. Tests will include overlapping speech, poor audio, network delay, obscured faces, and rapid editing. We will report accuracy, unavailable information, required corrections, processing time and cost, and differences across

recording conditions. Difficult cases will be retained rather than silently removed.

Measure validation. We will evaluate proposed measures separately from extraction accuracy. Correctly detecting a pause, pitch change, or smile does not establish what it means. For each measure, researchers will specify the construct, observations, baseline, combination rule, and treatment of unavailable information. We will test agreement with human coding or established measures, stability across reasonable measurement choices, added information beyond transcripts, and performance in new settings. Null findings and disagreements across channels will be retained.

Researcher use. We will develop a guided interface through which researchers can process a recording, inspect the source of an observation, correct assignments, specify and compare measures, and export the results. Researchers outside the project will reproduce the pilots and develop new measures, testing whether the system can be used independently.

Empirical expansion. For earnings calls, we will enlarge the executive panel, construct personal baselines using only earlier observations, and define outcome tests in advance. For video, we will study repeated interviews of the same executives and interviewers across programs and time. These studies will help establish when particular measures are informative and where their limits lie.

Stage 2 will produce a documented system released under appropriate open-source terms, a versioned framework for defining measures, a human-validated benchmark set, and a technical appendix reporting the system's performance and limits.

5. Risks and boundary conditions

The main risks are false precision, unsupported psychological interpretation, demographic and cultural bias, privacy, changes in the models used, and differences introduced by microphones, network delay, camera editing, or recording setting. We will address these risks by keeping observations linked to the original recording, reporting unavailable information and exclusions, comparing reasonable models and measurement choices, and using within-person or within-format comparisons when appropriate. Differences across people and recording conditions will be reported rather than averaged away.

The system will not infer emotion, deception, intent, personality, culture, or executive quality from behavioral information without a theory and independent evidence supporting that interpretation. Some constructs may not be observable from a recording, even when features such as pauses or facial movements are measured accurately. Establishing these limits is valuable because it prevents unsupported claims.

6. How organizational and social science would change

If successful, this project would make more organizational behavior research-observable. Recordings that are now reduced to transcripts or left unused could be studied as evidence of how people speak, respond, listen, influence one another, coordinate, and change across encounters. Researchers could examine moment-to-moment interaction across hundreds or thousands of recordings rather than only small, manually coded samples.

Because each measure remains connected to its underlying observations and the original recording, other researchers could inspect, revise, test, and adapt it. The goal is not faster transcription or automated psychological labeling. AI extracts and organizes information at scale; researchers define constructs, choose measures, validate interpretations, and remain responsible for inference. By reducing the cost of structuring recordings while preserving the source evidence, the system would make previously infeasible questions about organizational interaction possible to study.

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